

Notes on Radio Evolution of Galaxies

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Contents

1	Useful Terminology and Concepts	2
1.1	Basic Terminology	2
1.2	The Magnitude Scale	2
1.2.1	Apparent Magnitude	3
1.2.2	Filters and Colour	3
2	The World of Galaxies	5
2.1	Classification of Galaxies	6
2.1.1	Morphological Classification	6

Chapter 1

Useful Terminology and Concepts

1.1 Basic Terminology

- i. **Luminosity (L)**: The total amount of electromagnetic **energy emitted per unit time** by an astronomical object. L_B and needs to be mentioned when we bring up colour filters.
- ii. **Radiant Flux Density (S)**: The total amount of electromagnetic **energy per unit area per unit time** by an astronomical object. In other words, it is the luminosity per unit area. According to this definition, at a distance r from a star (say):

$$S = \frac{L}{4\pi r^2}$$

- iii. **Effective Temperature**:

1.2 The Magnitude Scale

Our eyes judge brightness according in a logarithmic fashion, similar to hearing. This means that what we perceive to be twice as bright is actually emitting a lot more than just twice the radiant energy. Due to this, optical astronomy uses ‘magnitude’ as a scale of brightness.

1.2.1 Apparent Magnitude

For two sources with fluxes¹ S_1 and S_2 , their *apparent magnitudes* m_1 and m_2 are defined as:

$$m_1 - m_2 = -2.5 \log \left(\frac{S_1}{S_2} \right)$$

This means that a brighter object will have a smaller apparent magnitude. The factor of 2.5 is chosen to most accurately match the historically calculated magnitudes.

1.2.2 Filters and Colour

Optical observations are performed with a combination of filters and detectors. Since the spectral energy distribution is not necessarily uniform over all wavelengths, apparent magnitudes are defined for each of these filters, i.e. apparent magnitude is a function of the wavelength of light being considered.

Vega Magnitude

$$m = m(\lambda) \equiv m_X \equiv X$$

where X is the filter for some specific wavelength λ .

The magnitudes of the different filters are related, by definition, as $U = B = V = R = I = \dots$ for main sequence stars of spectral type A0 (Vega is an archetype of such stars), i.e., every colour index for such a star is defined to be zero.

For a more precise definition, let $T_X(\nu)$ be the transmission curve of the filter-detector system. $T_X(\nu)$ specifies which fraction of the incoming photons with frequency are registered by the detector. The apparent magnitude of a source with spectral flux S_ν is then:

$$m_X = -2.5 \log \left(\frac{\int d\nu T_X(\nu) S_\nu}{\int d\nu T_X(\nu)} \right) + \text{const.}$$

The argument of the logarithm is the average flux received in total for the entire EM spectrum. The constant is determined from reference stars. This is also referred to as the Vega apparent magnitude.

¹Here, we are talking about radiant flux density, defined by energy emitted per unit area per unit time, i.e. luminosity per unit area.

AB Magnitude

Another commonly used definition, does not consider the spectral energy distribution of reference. The reference is instead defined as constant flux at all frequencies, $S_\nu^{\text{ref}} = S_\nu^{\text{AB}} = 2.89 \times 10^{-21} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}$. This value makes A0 stars like Vega have the same magnitude in the original Johnson-V-band as in the AB system.

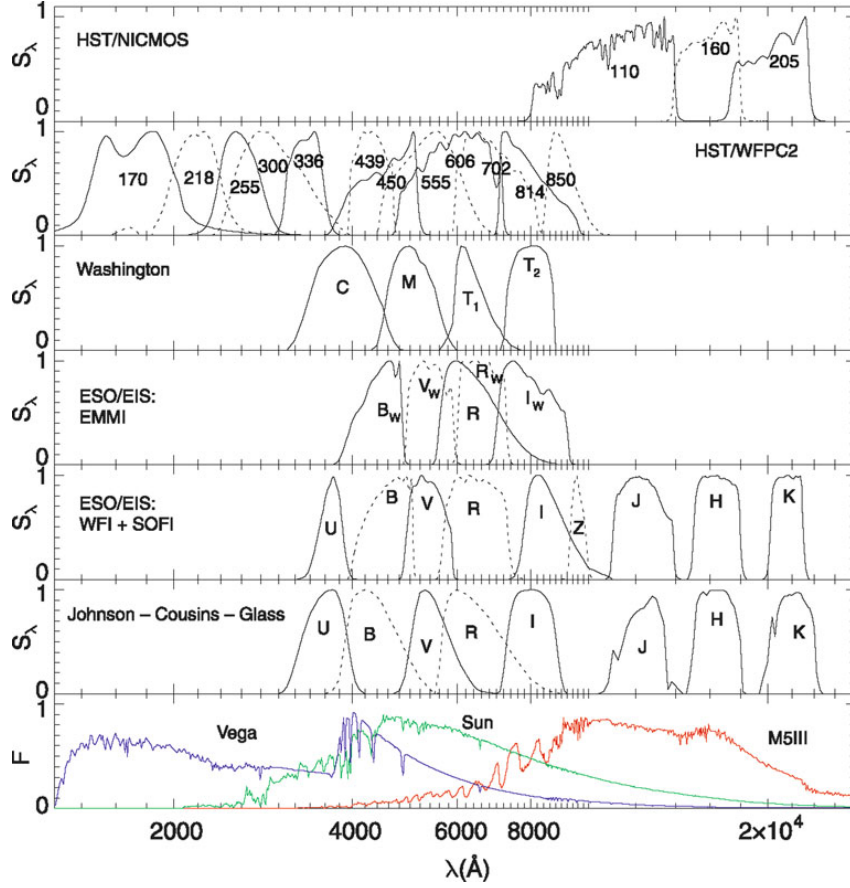


Figure 1.1: This shows the transmission curves of different spectra. At the bottom, we have the spectra of 3 stars with different effective temperatures.

Chapter 2

The World of Galaxies

Ever since the Hubble Deep Field image, the whole of humanity has set their sight to discover what lies far beyond they could ever comprehend. When dealing with extragalactic astronomy, the most notable entities are arguably other galaxies, and this is a chapter all about them.



Figure 2.1: The Hubble Deep Field image produced by *Edwin Hubble* changed the world forever. All the dots in the image are galaxies, not stars.

The light from 'normal galaxies' is mainly the sum of all the light emitted by the stars present within it. need to add note about age of stars and but for that I need to add the section about stars at the beginning.

2.1 Classification of Galaxies

Galaxies dafds

2.1.1 Morphological Classification

On the basis of shape or morphology, galaxies have been historically classified into the following categories:

1. **Elliptical Galaxies:** Their isophotes¹ resemble ellipses. They are further subclassified according to the **ellipticity** of the isophotes: $\epsilon = 1 - b/a$, where a , b are the semi-major and semi-minor axes. Then we define $N = 10\epsilon$ which gives us the representation EN for elliptical galaxies. An E7 galaxy is an elliptical galaxy with $b/a = 0.3$.
2. **Spiral Galaxies:** Their isophotes look like spirals.

¹Isophotes are contours along which the surface brightness is constant.